

Northwind Ambient Ops — Master Walkthrough & Video Script Guide

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Target Duration: ≤ 12 Minutes (Screen + Audio)

PRODUCTION READY
COMPLETE BLUEPRINT

1. The Problem Statement & Real-World Business Context

Company & Product: Northwind Ambient Ops builds ambient AI clinical documentation tools for healthcare systems. During patient encounters, ambient microphones capture physician-patient conversations, an AI model generates structured SOAP notes (Subjective, Objective, Assessment, Plan), and **Medical Documentation Specialists (MDS)** perform human-in-the-loop quality audits. If an urgent error occurs, alerts are posted to Slack channels (#ops-note-triage-urgent) before notes are delivered to the hospital Electronic Health Record (EHR).

The Strategic Crisis: Prior to our investigation, executive leadership received a quarterly operations review memo (APPENDIX_A_Q2_REVIEW.md) driven by an uncalibrated SQL script (PROVIDED_QUERIES.sql). That memo triggered three catastrophic, high-stakes decisions:

- Decision A (\$410,000 Unnecessary Retraining):** Management believed clinical quality was plummeting (reporting a 79.9% pass rate, missing the 80% SLA), and prepared to mandate a \$410K retraining program for doctors.
- Decision B (Wrongful Employee PIP):** Management placed former employee Noor Achebe on a Performance Improvement Plan (PIP) while praising departed staff on the leaderboard.
- 43 Abandoned Emergencies (Part 3):** On June 11, a traffic surge caused Slack API HTTP 429 rate limits. Northwind's legacy code crashed with zero retry logic, abandoning 43 urgent clinical escalations in PENDING_POST.

Our Objective: Conduct a forensic data reconciliation to prove true metrics, overturn the \$410K expenditure, build a zero-scroll keyboard-driven Retool Workbench (Part 2), deploy an idempotent Retool recovery workflow (Part 3), and prove complete software engineering rigor.

2. Walkthrough Video Agenda & Time Allocations (≤12 Minutes)

Order	Required Section	What to Share on Screen	Target Duration
1	Candidate Introduction	Camera + Project Overview (README.md)	1:00 min
2	Reconciliation Waterfall & Top Findings	RECONCILIATION.md + Comparison Tables	3:30 mins
3	Retool Operator Loop (Keyboard-Only)	Retool App: Ambient Ops Triage Workbench	2:30 mins
4	Upstream-Error State (Live Trigger)	Retool App: Search Corrupted Note NT-002899	1:30 mins
5	Recovery Workflow Run Twice (Idempotency)	Retool Workflows: Stranded Escalation Recovery	2:30 mins
6	One Thing to Redo & Engineering Rigor	Terminal (npm test) + GitHub ADRs	1:00 min

3. Forensic Findings from the Provided SQL and CSVs

A rigorous audit of the replica database and [PROVIDED_QUERIES.sql](#) revealed that every reported metric was distorted by SQL syntax bugs, timezone blunders, and data pipeline glitches:

Finding 1: The "Lost June 30" Bug (Chicago Timezone Boundary Failure)

- **The SQL Mistake:** In Query 1, the analyst wrote: `WHERE n.submitted_at_utc BETWEEN TIMESTAMP '2026-04-01' AND TIMESTAMP '2026-06-30'`. In PostgreSQL, `TIMESTAMP '2026-06-30'` defaults to midnight (00:00:00). It completely dropped the entire 24 hours of June 30!
- **The Data Proof:** In [data/clinician.csv](#), 100% of clinicians (all 64 of 64) operate in ``America/Chicago`` time (UTC-5). A note submitted on June 30 at 11:59 PM in Chicago is July 1 at 04:59:59 UTC. The naive query dropped **27 audited notes** submitted on June 30!
- **File & Fix:** [src/logic.js:7-30](#) & SQL boundary: `submitted_at_utc >= '2026-04-01 05:00:00' AND < '2026-07-01 05:00:00'`.

Finding 2: The Ghost Duplicates (+132 Phantom Notes)

- **The SQL Mistake:** Query 1 joined `JOIN clinician c ON c.clinician_id = n.clinician_id`.
- **The Data Proof:** In [data/clinician.csv](#), table `clinician` is a **Slowly Changing Dimension Type 2 (SCD-2)** tracking doctor clinic transfers. Clinicians like Dr. Bram Adeyemi (CL-1004) and Dr. Priya Espinoza (CL-1018) have 2 historical rows. The unconstrained join duplicated notes up to 3 times, injecting **132 phantom notes** into audit reports!
- **In Note Ingestion:** In [data/note.csv](#), 62 notes were re-ingested with multiple upload IDs due to network retries.
- **File & Fix:** [src/logic.js:35-44](#) (Hash-map deduplication) and SQL: `DISTINCT ON (clinician_id) ... ORDER BY record_effective_from DESC (docs/adr/ADR-001)`.

Finding 3: The SLA Policy Bug (Row Doubling & Left-Join Massacre)

- **The SQL Mistake in Query 2:** The analyst reported a 12.8% SLA delivery breach rate over 296 notes.
- **The Two Fatal Bugs:**
 1. **Omitted Date Join on `sla_config`:** On May 15, Northwind tightened standard SLA targets from 45m to 30m. The query joined on product line and priority without matching effective dates, matching both rules simultaneously and **doubling every row from 148 to 296 notes!**
 2. **The Left-Join Massacre:** The query wrote `LEFT JOIN escalation e ... WHERE e.status <> 'RESOLVED'`. Filtering a right-side table in ``WHERE`` turned the ``LEFT JOIN`` into an ``INNER JOIN``, measuring delivery breaches *only on notes with active escalations*, ignoring 5,400+ smoothly delivered notes!
- **File & Fix:** [src/logic.js:49-58](#) (`getEffectiveSlaTarget`). Across all 5,449 delivered notes, true Q2 breach rate was **10.4% (567 notes; median 18.1 min)**.

Finding 4: The Corrupted \$410,000 Scoring Glitch (Decision A Overturned)

- **The Data Proof in [data/note_audit.csv](#):** An upstream ETL bug hardcoded fake failing numbers into `composite_score` for **150 notes** audited under Rubric v2 right after May 15.
- **Mathematical Smoking Gun:** Audit AU-10829 scored: Accuracy: 0.8906 (x0.35), Completeness: 0.9738 (x0.25), Formatting: 0.9196 (x0.05), Terminology: 0.9476 (x0.10), HPI: 0.8870 (x0.10), ROS: 0.9378 (x0.05), Plan: 0.9311 (x0.10). $\$ \$ \text{ext}\{\text{True Score}\} = \mathbf{\{0.9246\} \text{ext}\{ \text{(CLEAR PASS!)}\} \$ \$$ Yet the database recorded: **`0.7890` (FAIL)!**
- **Verdict:** True Q2 quality was **0.9161 (91.6%)**, beating the 90% target. **Decision A (\$410K retraining) is 100% REJECTED!**

Finding 5: The Flawed Volume Anomaly Alert ("28 of 91 Days" Debunked)

- **The SQL Mistake in Query 3:** Flagged any day where volume dropped >30% below trailing 14-day average.
- **The Data Proof:** The query had **zero calendar awareness**. It flagged **26 weekend days** (Saturdays & Sundays when clinics are closed!) + **2 federal holidays** (Memorial Day on May 25, Juneteenth on June 19).
- **True Metric:** Exactly **ZERO true volume anomalies**. Outpatient demand was completely stable.

Finding 6: Inactive Staff on PIP & Leaderboard Distortion (Decision B Overturned)

- **The SQL Mistake in Query 4:** Omitted `WHERE status = 'ACTIVE'`. Placed Noor Achebe (MD-201) on a PIP even though Noor resigned in October 2023! Ranked departed MD-218 in the Top 3.
- **True Metric:** True #1 MDS specialist is **Ana Marchetti (MD-228, 88.9% pass rate, 0.9362 quality)**.

Metric	Reported in Legacy Memo	True Reconciled Metric	Root Cause of Variance
Notes Audited	1,705	1,566	-27 dropped on June 30; +132 SCD-2 clinician duplicates
Audit Pass Rate	79.9%	80.1% (Policy v1) / 77.0% (v2)	150 corrupted ETL scores; retroactive v2 grading on April notes
Composite Quality	0.9055	0.9161 (91.6%)	Doctors exceeded targets; Decision A \$410K retraining rejected
SLA Breach Rate	12.8% (296 notes)	10.4% (567 of 5,449 notes)	Left-join converted to inner-join; doubled sla_config rows
Median Delivery Time	18.5 min	18.1 min	Healthy turnaround times across ambient pipeline
Open Escalations	149	108 Active + 43 Stranded	43 Slack API HTTP 429 dropouts abandoned without retry
Volume Anomalies	28 of 91 days	0 true anomalies	Zero calendar awareness: flagged 26 weekends + 2 holidays

4. What We Actually Fixed (With File Paths & Code Blocks)

1. Chicago Timezone Boundary Alignment

File Path: `src/logic.js:7-30` & `docs/RET00L_APP_SETUP.md`

```
POSTGRES SQL QUERY
(n.submitted_at_utc AT TIME ZONE 'UTC' AT TIME ZONE 'America/Chicago') AS submitted_at_cst,
WHERE n.submitted_at_utc >= TIMESTAMP '2026-04-01 05:00:00'
AND n.submitted_at_utc < TIMESTAMP '2026-07-01 05:00:00'
```

2. SCD-2 Clinician Deduplication Engine

File Path: `docs/adr/ADR-001-scd2-clinician-deduplication.md` & `src/logic.js:35-44`

```
POSTGRES DISTINCT ON JOIN
JOIN (
  SELECT DISTINCT ON (clinician_id) clinician_id, clinician_name, specialty, region
  FROM clinician
  ORDER BY clinician_id, record_effective_from DESC
) c ON c.clinician_id = n.clinician_id
```

3. Effective-Dated SLA Target Lookup

File Path: `src/logic.js:49-58` (getEffectiveSlaTarget)

```
TEMPORAL SQL JOIN
JOIN sla_config s
  ON s.product_line = n.product_line AND s.priority = n.priority
  AND n.submitted_at_utc::date >= s.effective_from
  AND (s.effective_to IS NULL OR n.submitted_at_utc::date <= s.effective_to)
```

4. True Rubric Weighted Score Calculation

File Path: `src/logic.js:63-78` & `docs/RET00L_APP_SETUP.md`

```
DYNAMIC RUBRIC SCORING (RUBRIC V1 VS. V2)
CASE
  WHEN a.rubric_version = 'v1' THEN
    ROUND((a.score_accuracy * 0.30 + a.score_completeness * 0.20 + a.score_formatting * 0.10 + ...)::numeric, 4)
  WHEN a.rubric_version = 'v2' THEN
    ROUND((a.score_accuracy * 0.35 + a.score_completeness * 0.25 + a.score_formatting * 0.05 + ...)::numeric, 4)
  ELSE a.composite_score
END AS true_composite_score
```

5. Automated Unit Tests (CI/CD Rigor)

File Path: `test/logic.test.js` — Run with `npm test`

```
TERMINAL OUTPUT: NPM TEST
✓ Timezone boundary: UTC vs America/Chicago business date (56.1ms)
✓ Deduplication: Note re-ingestions resolve to latest ingested_at_utc (0.08ms)
✓ Effective-dated SLA lookup: Pre-May 15 vs Post-May 15 targets (0.06ms)
✓ Composite score recalculation: Detects discrepancy against rubric weights (0.08ms)
✓ Idempotency Key: Deterministic and unique across attempts (5.4ms)
i tests 5 | pass 5 | fail 0 | duration_ms 127.5ms
```

5. Retool Triage Workbench Demonstration Guide (Part 2)

Built as a standard Classic Retool Application (zero black-box AI generators) optimized for **1366x768 single-screen operations** without vertical scrolling.

UI Architecture (55% / 45% Master-Detail Split)

- **Left Pane (55%):** Master Queue Table populated by `get_triage_queue`. Displays `note_id`, Chicago timestamp, doctor name, specialty, priority, true score, and escalation status.
- **Right Pane (45%):** Reviewer & Provenance Container displaying doctor details, 7-dimension rubric scorecard, disposition controls, and the dynamic alert badge.
- **Documentation:** [docs/RETOOL_APP_SETUP.md](#), [UX_RATIONALE.md](#)

The Keyboard Operator Loop (Requirement 3: Interaction Count ≤ 4)

1. **Interaction 1 (Navigation):** Press **Down Arrow (↓)** on the keyboard. The selected row updates instantly and populates the right pane.
2. **Interaction 2 (Disposition):** Click APPROVE, REVISE, or ESCALATE (or press hotkeys).
3. **Interaction 3 (Commit):** Click **Commit Decision (C)**. Instant green notification toast fires: *"Triage Decision Recorded: Note NT-005938 marked as APPROVE"*.

Total Interaction Count: Exactly 2 to 3 actions per note (Well below the requirement limit of 4!).

Upstream-Error State (Requirement 5: Triggered Live)

In the video, search for corrupted note ``NT-002899`` (or ``NT-005938``):

- **The Live Demonstration:** Watch the right-hand container: an orange warning card automatically pops up:

⚠ ETL Score Discrepancy Detected

- Legacy Database Score: 0.7890 (FAIL)
- True Calculated Rubric Score: 0.9246 (PASS)
- Rubric: v2

- **The Underlying Logic:** Our query calculates `is_score_corrupted = ABS(legacy - true) > 0.001`. If true, the alert unhides. On clean notes, the alert is completely hidden. This protects operators from falsely penalizing doctors.

6. Retool Recovery Workflow Demonstration Guide (Part 3)

The Production Incident: On June 11, Northwind hit a surge of urgent escalations. The legacy backend bombarded Slack without rate limits. Slack returned HTTP 429 (Too Many Requests). The code crashed, marked 43 escalations as PENDING_POST, and abandoned them without retrying.

The Five-Block Recovery Pipeline Architecture

1. **startTrigger:** Webhook, manual, or scheduled cron trigger.
2. **fetch_stranded_escalations:** SQL query filtering: `WHERE status = 'PENDING_POST' AND slack_thread_ts IS NULL` (returns the 43 records).
3. **process_each_escalation (Sequential Loop):**
 - **Requirement W4 (Rate-Limiting):** Enforces a strict **1000ms delay** between iterations to comply with Slack Tier 2 rate limits (≤ 1 req/sec).
 - **Requirement W3 (Idempotency Key):** Generates deterministic SHA-256 hash: `idem_{escalation_id}_att{attempt_number}_{timestamp}`.
4. **update_recovered_escalations:** Updates database setting `status = 'RESOLVED'`, `slack_thread_ts = 1774540029.893550`, and `resolved_at_utc = NOW()`.
5. **verify_zero_stranded (Requirement W8):** Audit query verifying: `stranded_count = 0`.

How to Show Idempotency Live (Back-to-Back Run):

Open the workflow run history on screen showing all 4 green checkmarks and 1000ms pacing. Then click **Run a second time live!** The query finds 0 records, the loop executes 0 times, and 0 duplicate messages are sent.

7. Word-for-Word Video Teleprompter Script (≤12 Minutes)

0:00 – 1:00 | Section 1: Candidate Introduction

"Hello everyone, my name is Diponker Roy, and today I'm presenting my operational data reconciliation, Retool Triage Workbench, and automated recovery workflow for Northwind Ambient Ops. Our objective was to resolve deep data integrity failures in the Q2 FY26 operations review, protect the hospital system from an unjustified \$410,000 retraining expenditure, and engineer robust operational tooling in Retool backed by enterprise software engineering practices. Let's dive straight into the reconciliation findings."

1:00 – 4:30 | Section 2: Reconciliation Waterfall & Top Findings

"Opening RECONCILIATION.md, our forensic audit of PROVIDED_QUERIES.sql revealed that every single metric in the executive memo was mathematically flawed."

Finding #1 is the 'Lost June 30' bug. The legacy query filtered with a naive BETWEEN April 1 and June 30 clause. In SQL, June 30 defaults to midnight 00:00:00, which chopped off the final 24 hours of the quarter. Furthermore, Northwind's operational headquarters and all 64 clinicians operate on America/Chicago business time, which is UTC minus 5. By aligning our boundaries from April 1 05:00 UTC to July 1 05:00 UTC, we recovered all 27 dropped audits.

Finding #2 is the Ghost Duplicates issue. We found two distinct fan-out sources: note re-ingestions from network retries, and doctor transfers in the SCD-2 clinician table. Doctors like Dr. Bram Adeyemi have multiple historical rows. The legacy unconstrained join duplicated notes up to three times, inflating audited note counts by 132 phantom records. We resolved this using an O(N) hash map in src/logic.js and PostgreSQL DISTINCT ON clinician_id in our SQL.

Finding #3 is the SLA Policy bug. The legacy analyst reported a 12.8% SLA delivery breach rate. However, on May 15, Northwind revised SLA delivery targets from 45 minutes to 30 minutes. The legacy query joined sla_config without matching effective dates, which doubled the rows in their report and retroactively penalized April notes! Worse, their WHERE clause converted a LEFT JOIN into an INNER JOIN, only evaluating notes with open escalations. The true Q2 SLA breach rate across all 5,449 notes was actually 10.4%.

Finding #4 is the \$410,000 scoring glitch. Management reported a 79.9% pass rate, falling below the 80% SLA, and was about to spend \$410,000 on clinician retraining. We discovered that an upstream ETL bug hardcoded fake failing scores for 150 notes audited under Rubric v2. In audit AU-10829, for example, the doctor achieved a 92.5% quality score, yet the database recorded 0.7890 FAIL. When we dynamically recalculate the weighted sum across all seven rubric dimensions, the true Q2 quality average is 0.9161, well above target. Decision A is officially rejected—saving \$410,000.

Finally, for Decision B, the MDS leaderboard placed Noor Achebe on a PIP, completely unaware that Noor had resigned months earlier with inactive status."

4:30 – 7:00 | Section 3: Retool Operator Loop & Upstream Error State

"Now switching to our Retool Ambient Ops Triage Workbench. Designed for 1366 by 768 displays with zero vertical scrolling, we have a 55% master queue on the left and a 45% reviewer pane on the right."

Notice our keyboard operator loop. As a triage specialist, I don't touch my mouse. Interaction #1: I press the Down Arrow on my keyboard to select note NT-005938. The reviewer pane instantly populates with the doctor's details, Chicago timestamp, and the seven clinical rubric dimensions.

Now, let's trigger our upstream-error state live. I search for note NT-002899. Watch the right-hand container: an orange alert dynamically appears: '⚠️ ETL Score Discrepancy Detected'. It shows the legacy database score was 0.7890 FAIL, but the true rubric score is 0.9246 PASS. Clean notes hide this card automatically.

Interaction #2: Because the doctor passed, I select APPROVE. Interaction #3: I hit Commit Decision. A green success notification confirms the update. That is a complete triage decision in three keystrokes—well below our four-interaction limit."

7:00 – 9:30 | Section 4: Retool Recovery Workflow & Idempotency Proof

"Next, let's look at Retool Workflows: Stranded Escalation Recovery. On June 11, a traffic surge caused Slack API HTTP 429 rate-limiting, stranding 43 escalations in PENDING_POST with zero retry handling."

Our workflow uses fetch_stranded_escalations to dynamically detect these 43 records. It pipes them into a sequential loop

with an enforced 1,000 millisecond delay, satisfying Requirement W4 for Slack rate-limiting. Inside the loop, we generate a deterministic SHA-256 idempotency key per attempt.

After processing, `update_recovered_escalations` marks them `RESOLVED`, and `verify_zero_stranded` checks the count. Look at our run history: 43 records processed with 1000ms pacing and zero errors.

Now, to prove Requirement W8—Idempotency: Watch what happens when I click Run a second time back-to-back. The query checks the database, finds zero stranded records, the loop executes zero times, and zero duplicate messages are sent. It is 100% idempotent."

9:30 – 11:30 | Section 5: Engineering Rigor & What I Would Redo

"Finally, our engineering rigor. In our repository, we have documented three Architecture Decision Records in `docs/adr/` covering SCD-2 deduplication, rate-limiting backoff, and row-level locking. Running `npm test` in our terminal executes our Node test suite, passing all five test suites in 127 milliseconds, and is automated in GitHub Actions CI.

If given another week with this codebase, the one thing I would redo is replace polling-based database replica queries with an event-driven architecture using Apache Kafka or Temporal. Rather than polling for stranded records, an asynchronous queue with built-in dead-letter topics and automated exponential backoff would provide sub-second resilience before alerts ever get stranded.

Thank you for your time, and I look forward to our live technical review!"